Kalyan Reddy Katla

Houston, TX, 77407 | 937.986.9917 | kalyan.katla@gmail.com | LinkedIn | GitHub

Senior AI/ML Engineer with 8+ years of building large-scale production AI systems in fraud detection, risk intelligence, graph ML, and LLM-based agentic automation. Proven track record of shipping enterprise-grade ML systems combining transformers, generative AI, and graph-based learning to reduce operational cost, improve detection accuracy, and automate decision workflows at scale.

CORE EXPERTISE

LLMs, RAG Systems, Agentic AI (ReAct, Tool Calling), Graph ML, Fraud/Risk Modeling, Recommendation Systems, Distributed ML Systems, Tabular Transformers, Computer Vision, Anomaly Detection, GenAI and ML Evaluations, Fine-Tuning, ETL pipelines.

RESEARCH & INNOVATION

- **Tabular Transformer Foundation Models**: Conceptualized and built Transformer architectures optimized for tabular datasets to automate feature engineering and improve sample efficiency on high-volume data streams.
- **Compression-Based Similarity (NCD):** Introduced an information-theoretic, parameter-free similarity framework leveraging Normalized Compression Distance as an alternative to learned embeddings, yielding superior fraud detection metrics.

PROFESSIONAL EXPERIENCE

Microsoft (Contract) **Sep 2025 - Present**

Senior Applied AI Engineer ***Seattle, WA***

- Reduced fraud investigation workload by **60% ($2M+ savings)** via multi-agent AI triage system built on Azure AI Foundry
- Developed T**abular Transformer foundation model prototypes** to reduce manual feature engineering across enterprise
- Improved coordinated fraud detection by **20%** using **graph-based entity linking (10M+ nodes)** and **NCD similarity scoring**
- Built production **graph ML system** for coordinated abuse detection, identifying fraud rings without labeled supervision
- Developed hybrid ML system combining **graph, XGBoost, embedding-based similarity models** for fraud risk scoring
- Built automated **feature validation, drift detection, evaluation systems**, improving data reliability in production pipelines
- Built scalable **data pipelines using Databricks, Event Hubs, ADLS Gen2**, supporting daily refresh of multi-TB datasets
- Implemented **online inference pipeline with** low-latency scorin**g REST APIs**, enabling real-time fraud decisioning at scale
- Designed CI/CD system for ML models enabling automated **retraining, validation, and rollback capabilities**
- Established observability for ML systems including **latency, drift, and model performance monitoring dashboards**
- Developed **agentic AI evaluation pipeline for multi-step reasoning systems**, measuring tool-use accuracy, planning quality, and task success rate across production workloads
- Implemented **MLflow-based** experiment tracking, reproducible training and governed model deployments
- Partnered with **product, engineering, and risk analyst teams** to define ML system requirements, align on evaluation metrics, and translate business needs into production AI workflows

Chevron Corporation (Contract) **Apr 2019 - Aug 2025**

Data Scientist ***Houston, TX***

Enterprise Threat Intelligence & Risk AI Platform

- Built real-time **threat intelligence system ingesting OSINT, web, dark web, and proprietary feeds**, transforming unstructured multilingual data into structured intelligence, reducing manual triage effort by **35%**
- Fine-tuned **BERT-based multi-task NLP models (intent classification, NER, event attribute extraction)** to convert raw intelligence feeds into structured features, improving downstream **risk classification accuracy and detection speed**
- Built **multi-agent AI system with role-based agents (investigator, retriever, and validator)** using ReAct reasoning and tool-calling to autonomously investigate, enrich, and escalate high-risk events through multi-step decision workflows
- Built **RAG-based decision support system** retrieving historical incidents and policy documents to generate **grounded mitigation recommendations aligned with geo-specific compliance rules**
- Improved enterprise QA relevance by **30% using retrieval-augmented generation with vector search and reranking models**, enhancing precision of retrieved context for downstream applications
- Engineered **geo-aware policy routing system** mapping detected events to regional compliance rules, escalation workflows
- Led technical design discussions for **end-to-end AI system architecture**, defining model strategy, data requirements, and deployment approach across multi-team initiatives
- Introduced **explainable AI** layer for security decisions, generating audit-compliant reasoning traces that map model outputs to source evidence, policies, and historical incidents
- Partnered with **cross-functiona**l engineering, product, and security teams to align ML system design, evaluation metrics, and deployment strategy for enterprise threat intelligence workflows

Industrial IoT + Edge AI System

- Improved early leak detection accuracy by **40% using Faster R-CNN-based** computer vision models deployed across refinery camera networks for real-time anomaly detection
- Achieved **<500ms** end-to-end inference latency by building edge-optimized video analytics pipelines eliminating cloud dependency for critical safety events
- Built **multivariate time-series** anomaly detection system for predictive maintenance over high-frequency industrial sensor telemetry, enabling early identification of equipment failure patterns
- Designed **multi-modal sensor fusion** pipeline combining computer vision, IoT telemetry, and statistical signal processing models for equipment failure detection and root cause analysis
- Implemented full-stack **ML observability framework** tracking model drift, inference latency, ingestion lag, and data quality metrics, reducing incident detection time by 45%

SVM Technologies **May 2016 - Nov 2016 Software**
Data *Engineer/Analyst* ***Hyderabad, India***

- Built Python/SQL ETL pipelines to automate data ingestion, reducing manual reporting overhead by 20%.
- Optimized relational database schemas and query indexes, drastically reducing report generation latency.
- Operationalized predictive ML models into production workflows, improving categorization accuracy by 25%.
- Enhanced code reliability by establishing unit testing frameworks and documentation.

Nokia Technologies **June 2015 – Jul 2015**
Engineering Intern ***Chennai***

- Identified process gaps in the Thermal Gel Dispensing Machine and implemented Embedded C/Arduino enhancements to improve automation, consistency, and reliability.
- Developed Embedded C code on Arduino to optimize machine control logic and improve dispensing consistency

PERSONAL PROJECTS

Sintra - Autonomous Edge LLM Optimization Agent (Sintra)

- Built an **agentic AI system for hardware-aware LLM optimization**, leveraging ReAct-based reasoning and multi-agent orchestration to automatically design and evaluate model compression strategies (quantization, pruning, architecture tuning) for edge deployment constraints
  Tech: Python, Hugging Face, Transformers, LangGraph-style agent design, LLMs

LocalCowork - Local-First Agentic AI Execution System (LocalCowork)

- Developed a **fully local agentic AI assistant** enabling autonomous task execution using ReAct reasoning, tool-calling (shell, Python, web search), and sandboxed code execution for safe workflow automation without cloud dependency
  Tech: Python, JavaScript, FastAPI, Ollama, LLMs, Docker sandboxing

TECHNICAL SKILLS

Generative AI & LLMs	RAG, fine-tuning (LoRA/SFT), embeddings, evaluation frameworks, prompt engineering
Agentic AI	ReAct, tool-calling, multi-agent systems, LangGraph, LangChain, MCP, Semantic Kernal, ADK
ML Systems	Graph ML, tabular transformers, anomaly detection, time-series, CV, NLP
Frameworks & Libraries	PyTorch, TensorFlow, FastAPI, Flask, Pandas, NumPy, Hugging Face, React, Next.js.
Systems	Distributed ML pipelines, real-time inference, CI/CD for ML, observability, ETL
Cloud & Orchestration	Azure (Databricks, AKS, Event Hubs, Synapse, AI foundry), AWS, Kubernetes, Docker.
Languages	Python, SQL, PySpark, C/C++, TypeScript

EDUCATION

Wright State University **Dayton, Ohio, 2018**

Master of Science in Computer Science

Course Work: Machine Learning, Distributed Computing, Computer Architecture, Advanced Computer Networks, Cryptography Data Security, Advance Database Management